

TECHNICAL REPORT AND USER **MANUAL**

WHALE TRACKER PRO – WHALE, ICEBERG, WALL, AND CRYPTOCURRENCY FUTURES MONITORING

1. GENERAL APPLICATION DESCRIPTION

Whale Tracker Pro is a real-time analysis tool for cryptocurrency markets (supporting USDT pairs). Its main objective is to detect and visualize:

- Large whale operations (high-value buys/sells).
- Iceberg orders (large orders fragmented over time).
- Liquidity walls (support/resistance in the order book).
- Futures alerts: abnormal growth in Open Interest (OI) and extreme funding rates.
- Volume profile by price.
- Taker buy/sell statistics, volume delta, and buy/sell ratios.

The application features a web interface with tabs for multiple pairs, interactive charts, a live event feed, parameter configuration, customizable sounds, quiet hours, configuration profiles, Telegram integration, and CSV data export/import.

2. COMPLETE LIST OF FUNCTIONS

2.1. Real-Time Monitoring Functions

#	Function	Description
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1	Whale Detection	Identifies operations that exceed a threshold (fixed or dynamic via sigma). Displays alerts in the feed, top banner, plays a sound, and shows a bar chart.
2	Iceberg Detection	Detects the accumulation of small operations that add up to a large volume within a configurable time window. Ignores icebergs caused by whales to avoid duplicates.
3	Wall Detection	Scans the order book periodically and triggers an alert when there is a configurable amount of BTC at the same price level in bids or asks.
4	OI Spike Alert	Monitors futures Open Interest and triggers an alert if it grows more than a configurable percentage within a specific time window (e.g., 10% in 5 min).
5	Extreme Funding Alert	Detects funding rates above an absolute threshold (e.g., >1% or < -1%) and displays the direction (longs pay shorts or shorts pay longs).
6	Volume Profile	Displays a horizontal bar chart representing the volume traded at each price level (useful for identifying

		support/resistance zones).
7	Whale Chart	Vertical bar chart showing the USD value of each whale trade, color-coded by buy (green) or sell (red). Features horizontal scrolling and Y-axis auto-scaling.
8	Real-Time Feed	A chronological list of events (whales, icebergs, walls, OI spikes, extreme funding) with color badges and details on price, quantity, and value.

2.2. Statistics and Metrics Functions

#	Function	Description
9	Metrics Cards	Displays: whale buys (count and volume), whale sells, total whales, taker buy volume, taker sell volume, volume delta (with a sentiment bar), current Open Interest, and funding rate.
10	Cross-Pair Ranking	Comparative table of all active symbols showing: total whales, buys, sells, USD volume, buy/sell ratio, volume delta, and current price.

11	Comparison Chart	Grouped bar chart visually comparing whale buys and sells across different pairs.
12	Top Metrics	Highlighting cards for the pair with the most whales, the one with the highest whale volume, and the one with the best buy/sell ratio.
13	Auto-Refresh	Charts and statistics automatically refresh periodically (every 5-10 seconds).

2.3. Configuration and Personalization Functions

#	Function	Description
14	Global Parameters	Adjust Sigma (standard deviation), fixed USD limit, iceberg time window (ms), and wall size in BTC.
15	Configuration Profiles	Save, load, and delete profiles containing all custom parameters (including Telegram settings and sounds).
16	Light/Dark Theme	Interface with light and dark mode toggles, persistent across local storage and the backend.
17	Language Switch	Spanish and English options

		with dynamic translation of the entire interface (including tooltips and labels).
18	"Whales Only" Filter	A mode that hides icebergs and small operations, displaying only whales, walls, and futures alerts.
19	Quiet Hours	Set specific periods (with day-of-the-week selection) during which alert sounds are deactivated.
20	Customizable Sounds	Choose specific audio files (.wav/.mp3) for whales, icebergs, walls, OI spikes, and extreme funding; includes the option to upload your own sounds.
21	Telegram Integration	Configure your bot token and chat ID to receive real-time alerts directly on Telegram.
22	CSV Export/Import	Export a pair's trade history to a CSV file or load a CSV to replay historical events.

2.4. Interface and User Experience (UX) Functions

#	Function	Description
23	Dynamic Tabs	Each added pair gets its

		own tab with dedicated charts, feed, and stats. Tabs can be closed individually.
24	Popout Windows	Open the chart, feed, or volume profile in separate, independent windows for multi-monitor tracking. Windows sync in real time.
25	Horizontal Scroll	Slider control and buttons to navigate the whale chart when there are too many data bars.
26	Collapsible Sections	Each UI panel (chart, feed, volume profile) can be minimized or expanded, remembering its state via localStorage.
27	Floating Alert Banner	Displays a temporary pop-up message in the top-right corner detailing the latest whale, iceberg, OI spike, or extreme funding event.
28	Connection Indicator	A green/red status LED next to the pair's name indicating whether its WebSocket connection is currently active.
29	Dynamic Price Formatting	Automatically adjusts displayed decimal places based on the asset's price magnitude (e.g., 0.00001234 displays 6 decimals).

2.5. Maintenance and Support Functions

#	Function	Description
30	Update Checker	Checks for new versions on GitHub upon startup and displays a notification if an update is available.
31	Integrated Support Page	Located at the bottom of the Config tab, featuring the support email, GitHub repository link, and Telegram group.
32	Error Logging	Captures and handles errors gracefully in the console, preventing the user interface from crashing.
33	Data Persistence	Themes, language, quiet hours, whales-only mode, and sounds are saved locally and/or in the backend.

3. TECHNICAL SUPPORT AND CONTACTS

3.1. Official Support Channels

Channel	Details
Email Address	baxcripz1989@gmail.com
GitHub Repository	https://github.com/barnaby1989/whale-updates (Documentation & Issues)
Telegram Group	@WhaleTrackerPro (Community Support & Announcements)

3.2. Version Information

- **Current Version:** v1.0.1
- **Developer:** Claw & Code
- **License:** Private Use (redistribution without permission is strictly prohibited)

3.3. Technical Requirements

- **Backend:** Python 3.8+ with the following libraries installed: eel, websockets, requests, numpy, pandas.
- **Frontend:** Modern web browser (Chrome, Firefox, Edge) supporting Web Audio API for dynamic sound playback.
- **Connection:** Stable internet connection to stream data from exchange APIs (Binance preferred).
- **Hardware Resources:** Minimum 4GB RAM, 2 CPU cores.

3.4. Troubleshooting Common Issues

Issue	Potential Solution
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Whales are not appearing	Verify that the fixed USD limit is not set too high or that the sigma value is not too large. Try lowering the values.
Sounds are not playing	Ensure the "Enable Sounds" checkbox is ticked and that no quiet hours are active. Check if your browser allows media autoplay.
Backend connection error	Rerun python main.py and ensure that port 8000 is not being blocked or used by another application.
Tabs fail to open	Clear your browser cache and reload the application. Open developer tools (F12) to check for console errors.
Telegram alerts not sending	Double-check that the bot token and chat ID are accurate. Ensure the bot has been started (/start) and has permissions to send messages to the group/channel.

4. USER MANUAL FOR CLIENTS

4.1. Getting Started

1. **Installation:** (If using the executable) – Run WhaleTrackerPro.exe or execute the script via terminal using python main.py.
2. **Initial Setup:**
 - Navigate to the **Config** tab and adjust your parameters:
 - Sigma (σ): 2.0 (Standard baseline. Higher values mean fewer, but larger, whales are

detected).

- Fixed Limit (\$): 50,000 USD (Any trade above this value will instantly count as a whale).
- Iceberg (ms): 2000 ms (The time window used to accumulate fragmented iceberg orders).
- Wall (BTC): 50 BTC (The total volume required at a price level to look like a liquidity wall).
- Click **Update** to save settings.

3. Adding Trading Pairs:

- Click the **+** Add Pair button.
- Type the symbol in lowercase ending with "usdt" (e.g., btcusdt, solusdt, ethusdt).
- Confirm. A brand new tab will appear with its live chart and event feed.

4.2. Daily Operation

Asset Tab View (e.g., BTCUSDT)

- **Whale Chart:** Every bar represents a single whale trade. Green = buy, Red = sell. The height represents its USD value.
 - Use the horizontal slider or the < > buttons to navigate through older bars.
 - The Y-axis auto-scales, but you can lock an upper boundary using optional settings.
- **Feed:** A running list of recent activities. You can collapse this panel by clicking the – button next to the title.
- **Volume Profile:** Displays structural support and resistance by highlighting where the heaviest trading volume occurred.
- **Live Stats:** Quickly read total buys, sells, taker metrics, delta balance, current Open Interest, and funding data.

Comparison Tab (Comparison)

- Displays a live competitive ranking of all active pairs.
- You can sort the data table by clicking on column headers or using the sorting dropdown.
- The grouped bar chart allows you to visually identify which pairs are experiencing aggressive whale buying vs selling.
- Top metrics cards highlight current market volume and momentum leaders.

Additional Settings

- **Telegram:** Paste your custom bot token and chat ID, then click save to receive remote alerts.
- **Profiles:** Save your active parameters as a custom profile (e.g., "my_btc_config") to

reload different styles instantly.

- **Alert Sounds:** Select audio presets for each event type or upload your own preferred .wav or .mp3 tracks.
- **Quiet Hours:** Set specific schedules (e.g., 22:00 to 08:00) and choose specific days to mute alerts automatically.
- **Themes & Languages:** Toggle dark/light interface layouts and switch text languages seamlessly between English and Spanish.

4.3. Interpreting Alerts

Alert	Market Meaning	Suggested Action
Whale BUY	A massive buy order (> threshold) executed. Potential bullish momentum.	Watch for trend continuation or look to set tighter trailing stops.
Whale SELL	A massive sell order executed. Potential bearish pressure.	Consider securing profits or adjusting your risk exposure.
Iceberg Order	Large hidden orders being quietly filled. Indicates stealthy institutional accumulation or distribution.	Observe if the iceberg acts as an hidden floor (buying) or ceiling (selling).
SUPPORT Wall	Significant buy liquidity resting at a specific price level.	Potential bounce zone for short-term long entries.
RESISTANCE Wall	Significant sell liquidity resting at a specific price level.	Potential rejection zone to watch for overhead supply.
OI Spike	Sudden surge in Open Interest. Brand new capital	Anticipate upcoming high volatility or sudden

	entering leverage markets.	liquidations.
Positive Extreme Funding	Funding rate is unusually high; longs are paying shorts.	Market may be overbought; look out for potential long-squeezes/corrections.
Negative Extreme Funding	Funding rate is unusually low; shorts are paying longs.	Market may be oversold; look out for potential short-squeezes/rebounds.

4.4. Data Management and Exporting

- **Export CSV:** Click the Export CSV button at the bottom of any asset tab to download the entire logged history of whales and icebergs for that asset.
- **Load CSV:** Use the Load CSV function to import a previously saved historical file, allowing you to replay events for backtesting or strategy reviews.

4.5. Tips for Advanced Users

- Fine-tune **Sigma** based on asset volatility. For lower-liquidity pairs (small altcoins), use a lower sigma like 1.2 or 1.5. For heavy pairs like BTC/ETH, 2.0 or 2.5 works best.
- Use the **Fixed Limit** alongside sigma if you prefer an absolute floor value. Setting both too low will flood the feed with noise.
- Turn on **Whales Only** mode if you prefer a clean workspace focused exclusively on massive market changes, filtering out smaller iceberg anomalies.
- Leverage **Popouts** to isolate charts onto separate screens if you use a multi-monitor trading setup.
- Popout windows sync instantly; a whale order registered on the backend updates the popped-out elements with zero added latency.

4.6. Contact and Bug Reporting

If you encounter software anomalies or have feature requests, please reach out via:

- **Email:** baxcripz1989@gmail.com (Include screenshots and reproduction steps where possible).
- **GitHub Issues:** Open an active ticket directly at [whale-updates/issues](https://github.com/baxcripz/whale/issues).

- **Telegram:** Join @WhaleTrackerPro for real-time community support and version drop updates.

5. VERSION RELEASE NOTES (v1.0.1)

- **New:** Integrated Cross-Pair Comparison Chart.
- **New:** Horizontal Volume Profile structural overlay.
- **New:** Futures Open Interest spike and extreme funding rate alert modules.
- **Improvement:** Sub-millisecond state synchronization across popout windows.
- **Improvement:** Universal native dark and light theme options.
- **Fix:** Patched an anomaly to prevent duplicate iceberg tracking during matching whale blocks.
- **Fix:** Upgraded order book deep-scanning compatibility across all supported data feeds.

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